12. Write the python program for Tic Tac Toe game

Aim: To develop a two-player Tic-Tac-Toe game using Python.

Procedure:

1. Create a 3 × 3 game board.
2. Display the board.
3. Allow players X and O to enter their positions.
4. Validate each player's move.
5. Check for a winner or a draw.
6. Display the final game result.

File Edit Format Run Options Window Help

```
board = [' '] * 9
wins = [
    (0, 1, 2), (3, 4, 5), (6, 7, 8),
    (0, 3, 6), (1, 4, 7), (2, 5, 8),
    (0, 4, 8), (2, 4, 6)
]
def display():
    for i in range(0, 9, 3):
        print('|'.join(board[i:i+3]))
        if i < 6:
            print('-----')
player = 'X'
while True:
    display()
    try:
        pos = int(input(
            f"Player {player}, enter position (1-9): "
        )) - 1
    except ValueError:
        print("Enter a number.")
        continue
    if pos not in range(9) or board[pos] != ' ':
        print("Invalid move!")
        continue
    board[pos] = player
    if any(all(board[i] == player for i in line)
           for line in wins):
        display()
        print("Player", player, "wins!")
        break
    if ' ' not in board:
        display()
        print("Game Draw!")
        break
    player = 'O' if player == 'X' else 'X'
```

Output:

```
 |X|
-----
 | |
Player O, enter position (1-9): 8
 | |
-----
 |X|
-----
 |O|
Player X, enter position (1-9): 9
 | |
-----
 |X|
-----
 |O|X
Player O, enter position (1-9): 1
O| |
-----
 |X|
-----
 |O|X
Player X, enter position (1-9): 6
O| |
-----
 |X|X
-----
 |O|X
Player O, enter position (1-9): 4
O| |
-----
O|X|X
-----
 |O|X
Player X, enter position (1-9): 3
O| |X
-----
O|X|X
-----
 |O|X
Player X wins!
```

Result: Thus, the Tic-Tac-Toe game was implemented successfully using Python.